Akoizo Original Practice - Form 3

Division C | Boomilever | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Why do most teams aim the chain near 41 cm rather than 45 cm?

A. Less reach means a smaller moment and smaller member forces

B. The rules require exactly 41 cm

C. Longer reach lowers the score directly

D. The bucket hits the wall at 45 cm

Answer: _____

2. (2 points) A bonus-style structure breaks at 11,000 g. How is it scored?

A. On 11,000 g with no bonus and no violation

B. On 11,000 g plus the bonus

C. Participation points only

D. Tier 2 with a construction violation

Answer: _____

3. (2 points) What is the first tiebreaker if two teams have equal scores?

Unit: The test apparatus, setup, and loading procedure.
Worked-case data: Plan the six-minute window for a team that has practiced setup.
Separate investigation observations:
Run 1: Setup took 5 min; fast pour; failed at 9 kg with violent swing.
Run 2: Setup took 2 min; slow pour with stick tips; held 15 kg.
Inspection: Both structures were identical builds from the same wood batch.

A. Fastest setup

B. Highest load

C. Lowest structure mass

D. Closest load estimate

Answer: _____

4. (2 points) A 30 cm x 0.3 cm x 0.3 cm balsa stick has a mass of 0.405 g. What is its density?

A. 1.5 g/cm^3

B. 0.45 g/cm^3

C. 0.15 g/cm^3

D. 0.015 g/cm^3

Answer: _____

5. (4 points) Explain why the top chord can be much thinner than the compression strut.

Answer: _____
_____
_____

6. (4 points) Why is a reliable 7 g design better than a one-time 6 g success?

Answer: _____
_____
_____

7. (4 points) Describe one way to keep the Loading Block centered within 2.5 cm of the wall centerline.

Answer: _____
_____
_____

8. (6 points) Solve this case and explain your reasoning: Compare strut forces for a full 15 kg (147 N) at 42 cm reach, with contact at 15 cm versus 10 cm.

Answer: _____
_____
_____

9. (2 points) Define “Structural efficiency” in the context of Boomilever.

Answer: _____
_____
_____

10. (4 points) How should a team choose its Estimated Load Supported?

Unit: The test apparatus, setup, and loading procedure.
Worked-case data: Plan the six-minute window for a team that has practiced setup.
Separate investigation observations:
Run 1: Setup took 5 min; fast pour; failed at 9 kg with violent swing.
Run 2: Setup took 2 min; slow pour with stick tips; held 15 kg.
Inspection: Both structures were identical builds from the same wood batch.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A strut buckles at 120 N with an unbraced length of 30 cm. A brace is added at its midpoint. Estimate the new buckling load.

Answer: _____
_____
_____

12. (2 points) Define “Static equilibrium” in the context of Boomilever.

Answer: _____
_____
_____

13. (2 points) What should the team do next?

Supplied synthetic observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

A. Return to 8.2 g and stop testing

B. Repeat Build 6 to confirm it is reliable

C. Compete with Build 4 because it is lightest

Answer: _____

14. (2 points) Define “Shear” in the context of Boomilever.

Answer: _____
_____
_____

15. (2 points) Define “Hypotenuse” in the context of Boomilever.

Answer: _____
_____
_____

16. (2 points) Define “Force couple” in the context of Boomilever.

Answer: _____
_____
_____

17. (2 points) Define “Build log” in the context of Boomilever.

Answer: _____
_____
_____

18. (2 points) Define “Stabilizing sticks” in the context of Boomilever.

Answer: _____
_____
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